

Proposed Presentation Schedule

- Feb 19th: Nick S., Carter M., Kayla S., Kylie H., Rafael B.
- Feb 24th: Maddy C., Ruby F., Devin H., Lily R., Tyler R.
- March 3rd: Aaron G, Alena C., Emily M, Madi V., Yichen L.
- March 5th: Lipika C., Cole M., Neev S, Chen-Yu C., Sophie C.

Section 1 of the Science Justification Editing.

HST Cycle 27 proposal [15902](#): PI Dan Weisz

Which of the Two Proposals do you think got Funded?

- Version Elephant
- Version Giraffe

Tracing the 6-D Orbital and Formation History of the Complete M31 Satellite System HST Proposal 15902

Here, we propose to image the entire M31 satellite system in order to establish a second local anchor point for understanding an entire galactic ecosystem.

- Version Giraffe was funded – Cycle 27
- Version Elephant was rejected (1st quintile) in Cycle 26, after having been rejected in Cycle 25 (2nd quintile)
- Awarded: 244 orbits - Treasury Program

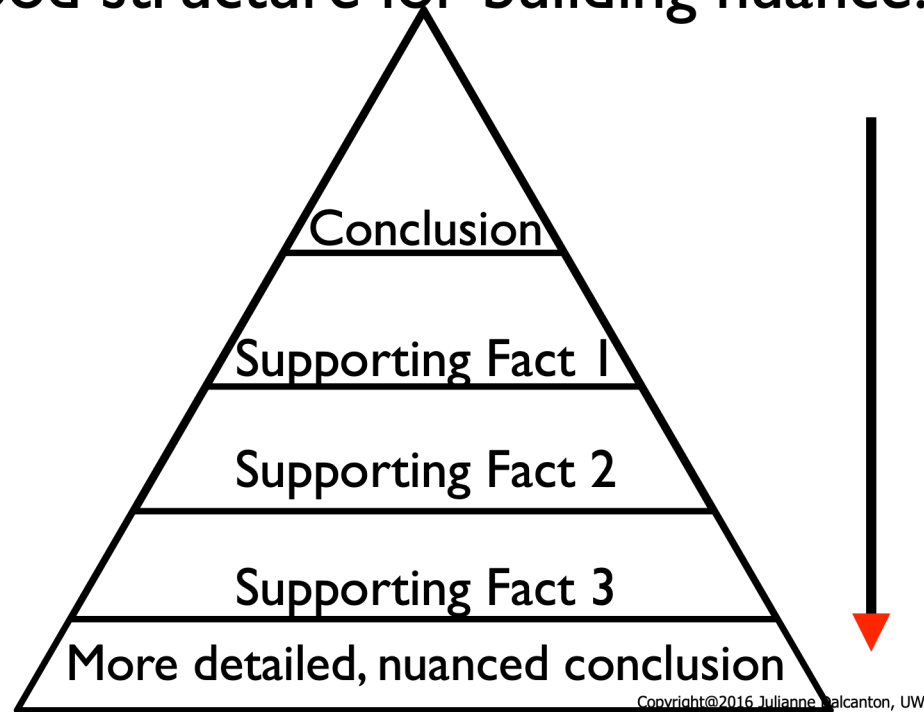
Q: What is the target?

Science Justification: Section 1

- 1-2 paragraphs: Facts
 - Explain/Justify “the Facts” stated in abstract. Aim for one “fact” per paragraph
 - Explain the importance of “the Fact” to the subfield at the end of each paragraph
- 1 paragraph: The Problem
 - Introduce the “Problem” – missing a “key component”
 - Explain why the “Problem” is important and *critical* to solve *now*.
- Last (1-2) paragraph(s): Synopsis
 - State the GOAL (“key component” with “target”) “We propose ...” . Underline this.
 - Justify why you’re applying for **this specific** instrument/facility/computing time
 - State a summary of the strategy to use the “key component” & “target”
 - State that the “key component” will fix “the problem”
 - Big picture statement/broader importance. (Legacy Value)

Paragraph Structure

Good structure for building nuance.



Opening Paragraph

HST Treasury Version Elephant

■ Scientific Justification

Our knowledge of low-mass galaxy evolution is anchored in the Milky Way (MW) halo. Within 300 kpc of the MW, HST observations of resolved stars have enabled the measurement of detailed star formation histories (SFHs) and precise proper motions (PMs) that complete the full 6-dimensional orbital phase space throughout the MW halo. These observations uniquely establish the relationship between dwarf galaxies and reionization (Brown et al. 2014; Weisz et al. 2014a; Boylan-Kolchin et al. 2015), provide stringent tests of CDM and the physics of galaxy formation (Boylan-Kolchin et al. 2012; Wetzel et al. 2016; Simon 2019), determine the 3-D orbits, infall times, and quenching timescales of several satellites (e.g., Kallivayalil et al. 2013; Sohn et al. 2013; Fritz et al. 2018; Simon 2018), and are sensitive to the internal motions and rotation of individual stellar orbits (van der Marel & Kallivayalil 2014). These results are fundamental to our understanding of all low-mass and satellite galaxies (e.g., Bullock & Boylan-Kolchin 2017).

Conclusion

More Detailed Conclusion

Q: what is the Target of the proposal?

■ Scientific Justification

Andromeda and its satellites are the next frontier for testing the Cold Dark Matter (CDM) paradigm and the physics governing the growth of galactic ecosystems across cosmic time. HST-enabled star formation histories (SFHs) and proper motions (PMs) of satellites around the Milky Way (MW) have established the relationship between dwarf galaxies and reionization (Brown et al. 2014; Weisz et al. 2014a; Boylan-Kolchin et al. 2015), and provided stringent tests of CDM and the physics of galaxy formation (Boylan-Kolchin et al. 2012; Wetzel et al. 2016; Simon 2019). PMs have been central to determining the 3-D orbits, infall times, and quenching timescales of several satellites (e.g., Kallivayalil et al. 2013; Sohn et al. 2013; Fritz et al. 2018; Simon 2018); and studying the internal motions and rotation of individual stellar orbits, themselves dictated by dark matter (van der Marel & Kallivayalil 2014). These results are fundamental to our understanding of all low-mass and satellite galaxies (e.g., Bullock & Boylan-Kolchin 2017). *This program will expand HST's role in these key areas of study to include M31 and its satellites, allowing us to comprehensively study an entire galactic ecosystem outside the MW halo for the first time.*

Conclusion

More Detailed Conclusion

Q: what is the Target of the proposal?

■ Scientific Justification

Elephant Version (not funded)

Where is the target???

Target is way too low in priority, leaving the reader wondering what the proposal is about for 3 paragraphs

Our knowledge of low-mass galaxy evolution is anchored in the Milky Way (MW) halo. Within 300 kpc of the MW, HST observations of resolved stars have enabled the measurement of detailed star formation histories (SFHs) and precise proper motions (PMs) that complete the full 6-dimensional orbital phase space throughout the MW halo. These observations uniquely establish the relationship between dwarf galaxies and reionization (Brown et al. 2014; Weisz et al. 2014a; Boylan-Kolchin et al. 2015), provide stringent tests of CDM and the physics of galaxy formation (Boylan-Kolchin et al. 2012; Wetzel et al. 2016; Simon 2019), determine the 3-D orbits, infall times, and quenching timescales of several satellites (e.g., Kallivayalil et al. 2013; Sohn et al. 2013; Fritz et al. 2018; Simon 2018), and are sensitive to the internal motions and rotation of individual stellar orbits (van der Marel & Kallivayalil 2014). These results are fundamental to our understanding of all low-mass and satellite galaxies (e.g., Bullock & Boylan-Kolchin 2017).

However, there is growing evidence that the MW satellites may not be broadly representative. Compared to the MW, satellite systems throughout the local Universe show varying luminosity functions, stellar populations, quenching properties, and spatial configurations, often in excess of cosmic variance (e.g., McConnachie & Irwin 2006; Brasseur et al. 2011; Tollerud et al. 2012; Chiboucas et al. 2013; Geha et al. 2017; Greco et al. 2018; Müller et al. 2018; Pawlowski 2018; Smercina et al. 2018). Thus, it remains unclear whether the fundamental insights established in MW satellites are generally applicable to low-mass systems or stem from the specific accretion history of the MW. Unfortunately, direct comparisons between the lifetime evolution of MW satellites and systems outside the Local Group (LG) are not currently possible. Even in the nearest groups (e.g., M81, Cen A; $D \sim 3.5$ Mpc), severe crowding and multi-decade time baselines limit the measurement of accurate SFHs and prohibit PMs, even with HST (and/or JWST). Exploring such distant systems in detail similar to the MW satellites requires a 12-16m space telescope.

In contrast, the M31 system provides a unique, yet under-utilized, opportunity to study an entire galactic ecosystem outside the MW halo. Current data hint at important differences in the internal (e.g., kinematics, stellar content; Collins et al. 2014; Martin et al. 2017) and global (e.g., “plane of satellites”; Ibata et al. 2013; Pawlowski 2018) properties of M31 and MW satellites.

Problem Paragraph

HST Treasury Version Elephant (not funded)

However, there is growing evidence that the MW satellites may not be broadly representative.

Compared to the MW, satellite systems throughout the local Universe show varying luminosity functions, stellar populations, quenching properties, and spatial configurations, often in excess of cosmic variance (e.g., [McConnachie & Irwin 2006](#); [Brasseur et al. 2011](#); [Tollerud et al. 2012](#); [Chiboucas et al. 2013](#); [Geha et al. 2017](#); [Greco et al. 2018](#); [Müller et al. 2018](#); [Pawlowski 2018](#); [Smercina et al. 2018](#)). Thus, it remains unclear whether the fundamental insights established in MW satellites are generally applicable to low-mass systems or stem from the specific accretion history of the MW. Unfortunately, direct comparisons between the lifetime evolution of MW satellites and systems outside the Local Group (LG) are not currently possible. Even in the nearest groups (e.g., M81, Cen A; $D \sim 3.5$ Mpc), severe crowding and multi-decade time baselines limit the measurement of accurate SFHs and prohibit PMs, even with HST (and/or JWST). Exploring such distant systems in detail similar to the MW satellites requires a 12-16m space telescope.

Conclusion – Problem Statement

More Detailed Conclusion

Problem Paragraph

HST Treasury Version Giraffe (funded)

There is growing evidence that the MW satellites may not be broadly representative. Compared to the MW, satellite systems throughout the local Universe show varying luminosity functions, stellar populations, quenching properties, and spatial configurations, often in excess of cosmic variance (e.g., [McConnachie & Irwin 2006](#); [Brasseur et al. 2011](#); [Tollerud et al. 2012](#); [Chiboucas et al. 2013](#); [Geha et al. 2017](#); [Greco et al. 2018](#); [Müller et al. 2018](#); [Pawlowski 2018](#); [Smercina et al. 2018](#); [Pawlowski et al. 2019](#)). Furthermore, current data suggest important differences in the internal (e.g., kinematics, stellar content; [Collins et al. 2014](#); [Martin et al. 2017](#)) and global (e.g., “plane of satellites”; [Ibata et al. 2013](#); [Pawlowski 2018](#)) properties of M31 and MW satellites. Thus, it is unclear whether the fundamental insights established in MW satellites are generally applicable to low-mass systems or stem from the specific accretion history of the MW.

Conclusion – Problem Statement

Supporting Fact Moved from another location in Elephant Version

More Detailed Conclusion – removed all the rest from the Elephant Version

Importance of the Problem/Urgency of Solution

HST Treasury Version Elephant (not funded)

history of the MW. Unfortunately, direct comparisons between the lifetime evolution of MW satellites and systems outside the Local Group (LG) are not currently possible. Even in the nearest groups (e.g., M81, Cen A; $D \sim 3.5$ Mpc), severe crowding and multi-decade time baselines limit the measurement of accurate SFHs and prohibit PMs, even with HST (and/or JWST). Exploring such distant systems in detail similar to the MW satellites requires a 12-16m space telescope.

In contrast, the M31 system provides a unique, yet under-utilized, opportunity to study an entire galactic ecosystem outside the MW halo. Current data hint at important differences in the internal (e.g., kinematics, stellar content; Collins et al. 2014; Martin et al. 2017) and global (e.g., “plane of satellites”; Ibata et al. 2013; Pawlowski 2018) properties of M31 and MW satellites.

Observationally, HST can measure PMs and SFHs at the distance of M31 to a precision comparable to MW satellites (e.g., Sohn et al. 2012; Skillman et al. 2017). However, there has been comparatively little investment in HST observations of M31 satellites. To date, > 1000 orbits have been dedicated to MW satellites, while only ~ 200 have been spent on M31 satellites. HST has observed nearly all 51 known MW satellites, while of the 35 known dwarf galaxies within the virial radius (~ 300 kpc) of M31, only 12 (34%) have HST imaging that reaches the oldest main sequence turnoff (MSTO) and only the 2 (6%) with recent star formation have published PMs (e.g., Brunthaler et al. 2007; van der Marel et al. 2018).

Conclusion – Importance/Urgency/Need Statement

Supporting Fact

What is this supporting? In Giraffe this is moved

Is this important to say here?? Detailed information about strategy

Importance of the Problem /Urgency of Solution

HST Treasury Version Giraffe (funded)

Direct comparisons between the lifetime evolution of MW satellites and systems outside the Local Group (LG) are not currently possible. Even in the nearest groups (e.g., M81, Cen A; D ~ 3.5 Mpc), severe crowding and multi-decade time baselines limit the measurement of accurate SFHs and prohibit PMs, even with HST (and/or JWST). Exploring such distant systems in detail similar to the MW satellites requires a 12-16m space telescope. M31 is thus the only logical target and it is now or never to start this survey. Measuring PMs with precisions similar to MW satellites (~ 40 km/s) at the typical distance of M31 (~ 800 kpc) requires an 8-12 year baseline with HST and JWST. PMs for M31 satellites cannot be measured with Gaia (stars are too faint and crowded) or AO (small field of view, insufficient PSF stability). It is imperative to establish first-epoch imaging now in order to leverage JWSTs full lifetime for optimal PM measurements.

Conclusion – Importance/Urgency/Need Statement

Supporting Fact

Supporting Fact 2 + Urgency – builds on previous

Much stronger urgency statement, moved up from later in the Elephant text

Synopsis Paragraph

HST Treasury Version Giraffe (funded)

Through this Treasury program, we propose a comprehensive ACS/WFC3 survey of the M31 satellite system, enabling unique and transformative HST science including critical issues in galaxy formation, cosmic reionization, and the nature of dark matter. Our imaging will target all known satellites within ~ 300 kpc of M31, encompassing its full halo profile out to the virial radius. It will have the depth to measure robust SFHs, the cadence to provide secure and precise distances using RR Lyrae, and the astrometric precision for optimal first-epoch proper-motion measurements, all comparable in quality to current measurements of the entire MW system.

HST Treasury Version Elephant (not funded)

Through this Treasury program, we propose a comprehensive ACS/WFC3 survey of the M31 satellite system. Our imaging will target all known satellites within ~ 300 kpc of M31, encompassing its full halo profile out to the virial radius. It will have the depth to measure robust SFHs, the cadence to provide secure RR Lyrae distances, and the astrometric precision for optimal first-epoch proper-motion measurements, all comparable in quality to current measurements of the entire MW

GOAL/the proposal – TO BE EXPANDED UPON

The Plan to use HST to solve the problem – TO BE EXPANDED UPON

Importance – TO BE EXPANDED UPON

Broader Impact – significance of the proposal – TO BE EXPANDED UPON

Cycle 25 TAC comments

- 2nd quintile

Strengths: The proposal was well-written and easy to understand. The proposal addresses a variety of compelling topics. The team is strong and experienced. The proposal makes good use of parallel observations and the data would have excellent archival value.

Weaknesses: By its nature, some of the science in this proposal will not be realized until the collection of second epoch astrometry in roughly a decade. Some panelists felt that the requested first epoch observations could be divided into multiple smaller requests over the next few years.

Cycle 26 TAC comments

- 1st quintile (so still in top , but not good enough)

Strengths: - The proposed program has the objective of obtaining accurate distances (from RR-Lyrae variables), detailed star formation histories (by reaching the ancient main sequence turnoff) for 23 M31 satellites (currently lacking appropriate data), as well as first epoch measurements for proper motion determinations (2nd epoch data in ~10yr from JWST) - The proposed program will provide (together with archival data) a complete census of the SFHs and (eventually) orbits of the 35 M31 satellites, to an accuracy comparable to MW satellites. This is a well described experiment with a straightforward path to get data.

Weaknesses: The panel felt that the scientific case was not compelling enough to be a large scale study, with some aspects of the outcome pending on future JWST use -- e.g. the PM depends on getting future JWST data. It was also not clear the extent to which RR Lyrae distance measurements might improve upon TPRGB measurements. Overall, the proposal seems better suited to a smaller pilot study, particularly given the substantial proposal pressure in the current cycle.

Cycle 27 TAC Comments

1st quintile and funded

- Strengths: This proposal aims to obtain star formation histories and distances of a complete sample of Andromeda dwarf galaxies, while also establishing an initial baseline for future proper motion measurements. These galaxies have not yet been observed with Hubble. The proposal is clear, compelling and well-written. The data would provide important constraints on the Andromeda system, and seed many different avenues of investigation (star formation histories, dynamics, simulations of the Andromeda system, etc.) This proposal would provide a powerful legacy for future studies in a number of different fields, including addressing pressing questions related to dwarf galaxies, their relationship to reionization, the stability of the 'great plane', the mass of Andromeda, and much more. The requested parallel observations were well justified.

Weaknesses: None noted.

JWST Cycle 3 proposal [15902](#): PI

Which of the Two Proposals do you think got Funded?

- Version Rabbit
- Version Javelina

COSMOS-3D: A Legacy Spectroscopic/Imaging Survey of the Early Universe

Scientific Category: Large Scale Structure of the Universe

Alternate Category: Supermassive Black Holes and Active Galaxies

Scientific Keywords: Cosmology, Emission Line Galaxies, Extragalactic Legacy And Deep Fields, Galaxy Clusters, Large-Scale Structure Of The Universe, Low-Luminosity Active Galactic Nuclei, Overdensities, Protoclusters, Reionization, Supermassive Black Holes

Instruments: MIRI, NIRCAM

Proposal Size: LARGE

Exclusive Access Period: 0 months

Allocation Information (in hours):

Science Time: 124.8

Coordinated Parallel Time: 120.8

Charged Time: 263.2

- Version Rabbit was funded (Cycle 3)
- Version Javelina was rejected in Cycle 2 (fifth quintile, triaged)

■ Scientific Justification Opening Paragraph Same for both proposals

COSMOS-3D: A Legacy Spectroscopic/Imaging Survey of the Early Universe

The *Hubble Space Telescope* has transformed our understanding of the Universe with extragalactic imaging surveys over the last three decades. There are two types of extragalactic

surveys: ultra-deep ‘pencil-beam’ surveys (e.g., HUDF, Williams et al. 1996), and wider but shallower surveys (e.g., COSMOS, Scoville et al. 2007). The ‘pencil-beam’ surveys enable the detection and characterization of faint galaxies. However, they lack the volume to probe the large-scale structure or detect the rarest galaxy populations. The Cosmic Evolution Survey (COSMOS) has allowed the first in-depth studies linking the formation and evolution of galaxies to their larger cosmic environments across cosmic time and substantially minimized uncertainties of key measurements arising from cosmic variance. It incorporates a wealth of multi-wavelength data, from radio to X-ray, to facilitate studies of galaxies and active galactic nuclei (AGN) over a wide range of energy and spatial scales. Leveraging the rich

multi-wavelength observations in the COSMOS field, the largest JWST Cycle-1 treasury program, COSMOS-Web (Casey et al. 2023), is adding multi-band infrared imaging in 1–5 μ m to this legacy field, which will extend our view of the Universe to an even earlier epoch, and bridge deep pencil-beam JWST surveys (e.g., JADES, Eisenstein et al. 2023) with shallower wide-area surveys by e.g., the future Roman Space Telescope (Akeson et al. 2019).

Conclusion – HST has been transformative in this subfield

More Detailed Conclusion – JWST is adding to this legacy – this is the future

Verion Javelina (not funded)

Problem Paragraph

Studies of the earliest galaxies and supermassive black holes (SMBHs) and the epoch of reionization (EoR) is one of the fundamental mission goals of JWST. Surveys such as COSMOS-Web can detect early galaxies photometrically, but can not accurately map their 3-D distributions or connect galaxies to the evolution of the intergalactic medium (IGM) in the EoR. NIRSpec/MSA spectroscopic surveys of deep fields can reach to faint objects at $z > 10$, but are limited to small survey volumes. The peak era of reionization is at $z = 6 - 10$; the typical ionizing bubble sizes and galaxy correlation lengths both demand surveys over scales of > 10 comoving Mpc for proper sampling. Many fundamental questions related to EoR can be addressed with a wide-field ($> 1000 \text{ arcmin}^2$) spectroscopic galaxy redshift survey focusing on this crucial redshift. From the JWST commissioning, the ERS and Cycle-1 observations, the slitless spectroscopic observations using NIRCам/WFSS mode (Greene et al. 2017) has been demonstrated to be the most powerful redshift survey machine for early galaxies (e.g., Sun et al. 2022; Kashino et al. 2022). It has the advantage of high survey speed and avoids the potential bias from photometric pre-selection, as shown by previous surveys such as 3D-HST (Brammer et al. 2012) and MUSE-Wide (Urrutia et al. 2019). The slitless mode is further aided by the realization that most young EoR galaxies have very strong rest-optical emission lines, allowing redshift measurements even for faint sources.

Conclusion – Problem Statement

The rest is facts about EoR without a real aim....

Big Picture Importance of the Problem

Verion Rabbit (funded)

Problem Paragraph

Studies of the earliest galaxies and supermassive black holes (SMBHs) and the epoch of reionization (EoR) are the most fundamental mission goals of JWST.

Surveys such as COSMOS-Web can detect early galaxies photometrically, but can not accurately map their 3D distributions. While NIRSpec/MSA spectroscopic surveys of deep fields can reach faint objects, their limited survey volume and shutter conflicts prevent us from revealing all the early cosmic structures.

To understand the emergence of early galaxies and SMBHs from the peak of their activities at $z \sim 2-3$ to the EoR at $z \sim 6-10$ (Planck 2020) through measurements such as galaxy correlation lengths and ionizing bubble sizes, we need surveys over scales of $\gg 10$ comoving Mpc for proper sampling. Many fundamental questions can only be addressed with a wide-field ($> 1000 \text{ arcmin}^2$) spectroscopic galaxy redshift survey charting the entire cosmic history.

Since the JWST commissioning, the NIR-Cam's wide-field slitless spectroscopy (WFSS) (Greene et al. 2017) has been demonstrated to be the *most powerful* redshift survey machine for the early Universe (e.g., Matthee et al. 2023a; Oesch et al. 2023) with its high survey speed and free from photometric pre-selection enabling us to reveal the whole structures from the EoR to the present epoch.

Conclusion – Problem Statement

The whole thing is shorter – the rest of the text supports the need for a solution and stresses importance

Big Picture Importance of the Problem – now in bold face – more like a subheading

Nuanced Conc: NIRCам as the answer.

Synopsis Paragraph - Rabbit

A large legacy wide-field spectroscopic survey is, thus, what is critically needed. We propose to perform a slitless spectroscopic survey ‘COSMOS-3D’ with NIRCam/WFSS targeting the COSMOS-Web field together with deep parallel MIRI imaging.

This will establish a treasury dataset covering the largest ever field (0.33 deg^2) in the infrared, with secure grism redshifts, enabling countless scientific investigations by the community for the next decade.

The survey uses WFSS F444W ($\sim 3.9\text{--}5.0 \mu\text{m}$) to cover the key rest-frame optical and infrared emission lines of early systems ranging from strong PAH features at $z \sim 0.2\text{--}0.5$, to line emission of $\text{Pa}\alpha$ and $\text{Pa}\beta$ at $z \sim 1\text{--}3$, He I and $[\text{S II}]$ at $z \sim 2.5\text{--}4.5$, $\text{H}\alpha$ and $[\text{N II}]$ at $z \sim 5\text{--}6.5$, up to $[\text{O III}]$ and $\text{H}\beta$ for $z \sim 7\text{--}9$, along with MIRI F1000W+F2100W to extend the deep mid-infrared ($\geq 10 \mu\text{m}$) coverage, critical to unambiguously determine the nature of early AGN demographics via their hot dust emission. The large MIRI coverage (482 arcmin^2) in the early JWST operation is urgent due to its declining sensitivity. The simultaneous F115W and F200W observations complete the full NIRCam filter ($\sim 1\text{--}5 \mu\text{m}$) coverage of the field, as well as adding the time-domain dimension for the studies of the earliest SMBHs.

This program will provide a panoramic view of the early Universe and address the following key scientific questions:

- 1. How did the early massive galaxies emerge? Do they pose a challenge to modern cosmology?*
- 2. How did the early SMBHs emerge? What is the nature of the puzzling ‘Little Red Dots’ unveiled by JWST?*
- 3. How did the entire ‘Cosmic Web’ of the Universe emerge? How did the Universe reionize? How did galaxies and SMBHs evolve in that context?*

**GOAL/the proposal –
TO BE EXPANDED
UPON**

**Broader Impact –
significance of the
proposal – TO BE
EXPANDED UPON**

**The Plan to use
JWST to solve the
problem – TO BE
EXPANDED UPON**

**Importance – TO BE
EXPANDED UPON –
this is missing in
Javelina – clear
deliverables.**

Weaknesses (TAC Report) – Version Javelina

The plan to conduct the observations in two medium bands (F360M and F480M) using only Grism R is perceived as a significant risk to confusion, especially for the densest regions where quiescent galaxies tend to be more clustered, and the proposal does not present a clear case against this. A more quantitative analysis would have been useful in this case, especially given the proposal's focus on robustly identifying protoclusters. While the need for the MIRI photometry for the AGN scientific case is clear, **the need for the NIRCam photometry, especially in the F115W, is less motivated**. It is unclear how the mentioned MUSE-wide data will be used in the analysis and will help strengthen the results. **While the observing methods and the sample construction techniques are reasonably well described, the actual scientific conclusions that one can draw as for the galaxy formation and evolution are not clear**. For example, by knowing M_{halo} , M_{star} , luminosity function, HI and galaxy distribution, and so on, it is not clear what models or scenarios will be tested, and to what end. **The proposal's focus seems to be disproportionally on methodology, rather than science-driven**. Redshifts are largely obtained with emission lines (star forming galaxies and AGN), and quiescent galaxies will be much less sampled. Therefore, the large scale structures will be basically traced by star forming galaxies which may not represent the overall structures including quiescent galaxies especially the densest regions where quiescent galaxies tend to be more clustered. This is perceived as a risk that is not recognized or discussed. The assertion that there will be "negligible impact from cosmic web" in the 0.33deg² NIRCam or 0.1deg² MIRI observations is not demonstrated in the proposal.

They lost the narrative by focusing on the methods

Note on Bold Face as Emphasis

Sand, JWST Cycle 2, Rejected

■ Scientific Justification

We request 23.6 (29.5 total) hrs of JWST+NIRCam imaging of Tucana B, a newly discovered, isolated ultra-faint dwarf (UFD) galaxy beyond the edge of the Local Group ($M_V \approx -6.9$ mag; $D \approx 1.4$ Mpc). Tucana B is a unique UFD, as it has likely never had an encounter with the Milky Way's or M31's hot halo or tidal field, yet its ground-based observations only display an old (~ 13 Gyr), metal poor ($[Fe/H] \sim -2$) stellar population, and it lacks an apparent neutral gas (HI) reservoir – it is ‘quenched’. Reionization is expected to quench many of the smallest galaxies at early times, heating their gas reservoirs and stopping any subsequent star formation. Observations of Tucana B down to the oldest main sequence turnoff (oMSTO) will be a definitive test of the role that reionization plays, identifying whether early star formation stopped around the reionization epoch or continued up until more recent times. Only JWST can perform these observations well outside the Milky Way's virial radius (within which UFDs have been affected by ram pressure and tidal forces) and Tucana B is currently the only target that can test this pillar of modern galaxy formation theory.

Motivation The faint end of the galaxy luminosity function is important for understanding dark matter and astrophysics on small scales (see, e.g. Bullock+17; Simon+19 for recent reviews).

Note on Bold Face as Emphasis

Sand, JWST Cycle 3, Accepted

■ Scientific Justification

We request 20.8 (26.1 total) hrs of JWST+NIRCam imaging of Tucana B, a newly discovered, isolated ultra-faint dwarf (UFD) galaxy beyond the edge of the Local Group ($M_V \approx -6.9$ mag; $D \approx 1.4$ Mpc). Tucana B is a unique UFD, as it has likely never had an encounter with the Milky Way's (or M31's) hot halo or tidal field, yet its ground-based observations only display an old (~ 13 Gyr), metal poor ($[\text{Fe}/\text{H}] \sim -2$) stellar population, and it lacks an apparent neutral gas (HI) reservoir – it is ‘quenched’. Reionization is expected to quench many of the smallest galaxies at early times, heating their gas reservoirs and stopping any subsequent star formation. Observations of Tucana B down to the oldest main sequence turnoff (oMSTO) will be a definitive test of the role that reionization plays, identifying whether early star formation stopped around the reionization epoch or continued up until more recent times. **Only JWST can perform these observations well outside the Milky Way's virial radius (within which UFDs have been affected by ram pressure and tidal forces) and Tucana B is currently the only target that can test this pillar of modern galaxy formation theory.**

Motivation The faint end of the galaxy luminosity function is important for understanding dark matter and astrophysics on small scales (see, e.g. Bullock+17, Simon+19 for recent reviews).

Applying this format to Synopsis for Longer Proposals

- NSF CAREER. G. Besla – first short section leading the proposal

Facts/Imp	1. Intellectual Merit: Since pioneers like Dr. Vera Rubin first discovered that stars in galaxies are moving much faster than expected based on a galaxy's visible mass budget, a huge endeavor has been undertaken across particle physics to astronomy disciplines to understand the nature of this unseen
Problem	mass. Despite these concerted efforts, the direct detection of "dark matter" remains elusive. The
Goal	research goal of my CAREER project is to develop novel methods and simulations to quantify the distribution and nature of dark matter about the Milky Way using high-precision measurements of the kinematics of nearby astrophysical objects such as galaxies, stars, stellar streams, and globular clusters.
Strategy	My CAREER project will calibrate the properties of the dark matter wake induced by the Large Magellanic Cloud (LMC) under different assumptions of the dark matter particle's nature—cold dark matter (CDM) vs. fuzzy dark matter (FDM) and self-interacting dark matter (SIDM). My group and I will further use a basis field expansion technique (Self-Consistent Field Method; SCF) to analytically quantify the CDM dark matter distribution of the Milky Way+LMC. We will validate this technique using cosmological zooms of analogous Milky Way + massive satellite systems in the <i>Auriga</i> CDM cosmological simulations. With these simulations and analytic realizations, we will establish a new
Importance of solution	framework for current/upcoming high-precision kinematic data of halo tracers (satellites, stellar streams, globular clusters, and halo stars) and tackle three fundamental questions in astrophysics:
Broader Impacts	1) Can we observe dynamical friction in action and what might this tell us about the nature of the dark matter particle? 2) What is the distribution of dark matter in the Milky Way in CDM? 3) What is the virial mass of the Milky Way ?